Secure By Design

EDUCATE. EMPOWER. SECURE THE FUTURE.

COURSE 1.1 / COMPANION WORKBOOK

Windows Server 2025

DC01 and DC02

Build the identity foundation

A practical companion for configuring Active Directory on two domain controllers, checking the result, and recording evidence.

MATERIALS EDITION 0.5 / COURSE 1.1

Updated 9 October 2026 · sbd-dc01-dc02

NAME _____ / DATE _____

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Table of contents

Understand the services, follow the configuration, and keep evidence of what you observe. Select a title to jump to that page.

What is Active Directory? 3

Why use two domain controllers? 4

What crosses the network? 5

Plan your build 6

Build DC01 7

Add DC02 8

Verify the domain 9

Keep a useful build record 10

Continue the learning path 11

Notices and permissions 12

Use it alongside the course

The interactive companion adds chapter progress, knowledge checks, evidence notes, and a 3D view of the network as it takes shape. The workbook is your readable reference and writing space.

YOUR PACE Read the foundations first if these concepts are new. Return to them whenever a configuration choice needs a clearer explanation.

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What is Active Directory?

The shared directory behind domain identities.

A directory for people and computers

Active Directory Domain Services (AD DS) is the Windows Server directory service. It stores objects such as user accounts, computer accounts, and groups, together with their attributes. Administrators use this shared directory to organize identities and manage access consistently.

The domain and the controller

A domain is a logical collection of directory objects, named here sbd.test. A domain controller (DC) is a server running AD DS that stores and serves that directory. DC01 and DC02 are two servers serving the same domain. A forest contains one or more domains; this build starts with one.

Identity, permissions, and policy

Authentication checks who is signing in. Authorization determines what that identity may access. Signing in successfully does not grant access to every resource: the destination service still applies its permissions. Groups make it practical to assign access to roles instead of managing every person separately.

Organization and consistent settings

Organizational units (OUs) organize objects and support delegated administration and Group Policy. Group Policy applies centrally managed settings to users and computers. A global catalog helps find directory objects across a forest; both controllers in this course provide that role.

REMEMBER AD DS stores the directory, and every domain controller serves it. Authentication proves who someone is; authorization decides what they may reach.

THINK IT THROUGH Explain the difference between the sbd.test domain and the SBD-DC01 server.

Microsoft · What is Active Directory Domain Services?
Microsoft · Group Policy and organizational units
Microsoft · Authentication and authorization

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Why use two domain controllers?

Keep the directory available when one server is unavailable.

Remove the single-server dependency

With one DC, its outage removes the only live directory service point. With two healthy, reachable controllers, clients can use a surviving DC for supported operations. This also lets you patch or restart one server while the other continues serving requests. Reconnection may take time; uninterrupted service is not guaranteed.

Both controllers do useful work

Both writable DCs can process normal directory changes and authentication requests. Replication shares directory changes between them. They are not a primary server and an idle spare. Certain operations still depend on a single FSMO role holder, so role placement and recovery need their own plan.

Make the redundancy real

Microsoft recommends at least two virtualized DCs per domain on different virtualization hosts. Two VMs on one host share that host as a failure point. Consider shared storage, power, and network dependencies too.

Availability still needs recovery

Replication can spread an accidental deletion or harmful change to both servers. Keep protected backups and test recovery separately. Two is a starting point for this design, not a universal capacity target: additional sites, users, and recovery needs may require more controllers.

REMEMBER Before relying on DC02, verify replication, review shared dependencies, and confirm both controllers pass their health checks.

THINK IT THROUGH If both controllers run on the same physical host, which outage would still stop both services?

SEE IT INSIDE THE NETWORK

DC01 is down, step 3: with SBD-DC01 offline, the workstation authenticates with SBD-DC02 instead, so sign-in continues.

securebydesigninc.com/inside-the-network#process=dc-down&step=3

Microsoft · Avoiding single points of failure
Microsoft · Operations master role placement

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What crosses the network?

The ports a domain member needs to reach its domain controllers.

Your lab is one flat network, so nothing stands between a client and DC01. In a segmented network, every one of these conversations has to be allowed on purpose. Inside the Network shows them crossing the core switch. The table starts after name resolution, which course 1.2 covers.

PURPOSE	PORT	WHAT IT CARRIES
Authentication	TCP/UDP 88	Kerberos tickets
Password change	TCP/UDP 464	Kerberos password changes and resets
Directory	TCP/UDP 389 · TCP 636	LDAP and LDAPS; UDP 389 carries the domain controller locator ping
Global catalog	TCP 3268 · 3269	Forest-wide searches; 3269 uses TLS
Group Policy and SYSVOL	TCP 445	SMB reads of policies and scripts from domain controller shares
RPC services	TCP 135 · 49152–65535	Endpoint mapper, then the dynamic range for NetLogon, LSA, and SAM
Time	UDP 123	W32Time; members follow the domain time hierarchy
Administration	TCP 9389	AD Web Services for PowerShell and Administrative Center, from administration hosts only

REMEMBER Allowing Kerberos isn't enough. Applying Group Policy and reading SYSVOL also need LDAP, SMB, and RPC. In the Inside the Network design, rule F03 permits this directory profile for domain members, with LDAP signing and channel binding required.

THINK IT THROUGH A workstation gets Kerberos tickets, but Group Policy never applies. Which ports would you check first?



SEE IT INSIDE THE NETWORK

Sign in, step 4: after it authenticates, the workstation reads directory information and Group Policy from SBD-DC01.

securebydesigninc.com/inside-the-network#process=signin&step=4

Microsoft · Service overview and network port requirements for Windows
Microsoft · Configure a firewall for Active Directory domains and trusts

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01 / PREPARE

Plan your build

Build two new domain controllers in one isolated network. The interactive guide contains the detailed steps, verification commands, and chapter evidence fields.

What you will learn

- Configure DC01 and add DC02 to the same domain.
- Inspect service state, replication health, and both controllers' roles.
- Connect configuration choices to security reasoning and evidence.

Before you begin

Use two new Windows Server 2025 Desktop Experience VMs, console access, and approved installation media. Record your starting OS builds. Keep this exercise isolated; sample domain values must match your own environment plan.

Sample configuration

SETTING	SAMPLE VALUE
Domain and NetBIOS	sbd.test / SBD
First controller	SBD-DC01 / 10.0.10.10/24
Second controller	SBD-DC02 / 10.0.10.11/24
DC02 DNS server	10.0.10.10 (SBD-DC01)
Network and time	Isolated subnet / record the internal time source

CM-2 BASELINE Record the intended values and the observed starting configuration. Keep the baseline aligned with changes to the build.

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Build DC01

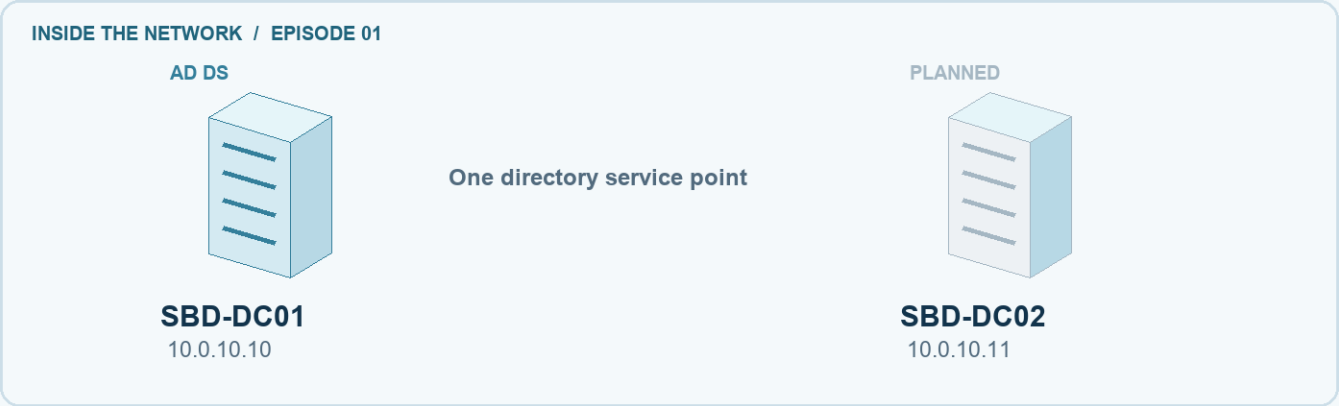
WHY THIS MATTERS Create the sample forest and its first domain controller.

Follow the build

1. On DC01, add the Active Directory Domain Services role and its management tools through Server Manager.

2. Promote the server using Add a new forest and sbd.test. Leave DNS server selected; the DNS delegation warning is expected, and course 1.2 explains both. The first controller is always a global catalog. Choose Windows Server 2025 functional levels and a unique DSRM password.

3. Review the SBD name, paths, and prerequisites. Read any warnings, resolve blocking errors, install, and restart.



NETWORK STATE **DC01 configured; DC02 planned.**

Check your result

```
Get-ADDomain | Select-Object DNSRoot, NetBIOSName
Get-Service NTDS,Netlogon
Get-SmbShare -Name SYSVOL,NETLOGON
```

Expect the planned domain and running services, with both shares present. Investigate missing results before adding DC02.

CM-6 SETTINGS Record the promotion choices and why they fit the environment. Keep passwords out of evidence.

COMMON MISTAKE Role installation alone does not complete domain-controller promotion.

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Add DC02

WHY THIS MATTERS Add another writable controller to the same domain.

Follow the build

1. Point DC02 at DC01 (10.0.10.10) for DNS and leave it that way; course 1.2 explains the DNS design. Check network reachability and clock alignment.

2. Install AD DS, then choose Add a domain controller to an existing domain. Use sbd.test and authorized domain credentials. Leave DNS server and global catalog selected.

3. Review site, replication source, DSRM password, and prerequisites. Install and restart.



NETWORK STATE Two controllers with a replication relationship.

Check your result

```
Get-ADDomainController -Filter *
repadmin /replsummary
repadmin /showrepl SBD-DC02
```

Expect both controllers and successful recent replication. Record any unresolved errors.

CP-9 RECOVERY A second controller supports availability. Backups and tested recovery remain separate work.

COMMON MISTAKE Choose the existing-domain option. A new forest would create a separate directory.

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Verify the domain

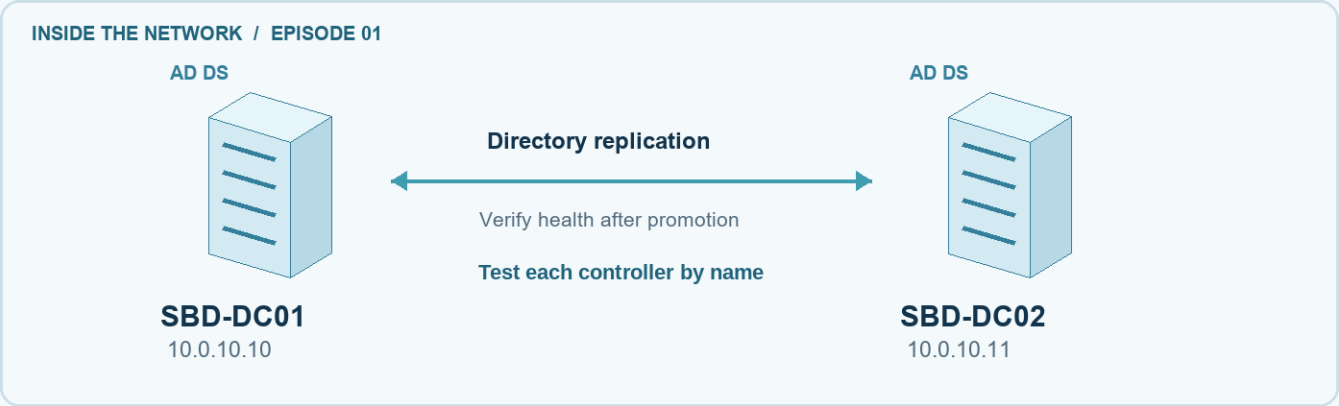
WHY THIS MATTERS Ask each controller for an observable result.

Follow the build

1. In an elevated PowerShell window on either controller, run DCDiag against SBD-DC01 and then SBD-DC02. Read each test's result, not just the last line.

2. Check replication both ways: the summary, then each controller's inbound partners. Expect recent successful attempts in both directions and no failures.

3. List both controllers' global catalog status and the operations master role holders; SBD-DC01 holds all five. Confirm SYSVOL and NETLOGON are shared on SBD-DC02 too.



NETWORK STATE Check each controller and confirm replication both ways.

Check your result

```
dcdiag /s:SBD-DC01
dcdiag /s:SBD-DC02
repadmin /replsummary
repadmin /showrepl *
Get-ADDomainController -Filter * | Select-Object HostName, IsGlobalCatalog
netdom query fsmo
Get-SmbShare -Name SYSVOL,NETLOGON -CimSession SBD-DC02
```

Expect each DCDiag test to pass or have a recorded cause, successful replication both ways, and SYSVOL and NETLOGON on both controllers. A command that ran is not a pass.

IA-2 SIGN-IN Healthy, replicating controllers are a precondition for domain sign-in. These checks do not establish all authentication requirements.

COMMON MISTAKE A successful ping does not show replication, SYSVOL, or sign-in health.

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05 / EVIDENCE

Keep a useful build record

Use My evidence in the interactive guide to save chapter notes, confirm your checks, and answer each question. Export your record before moving to another browser or device.

Your evidence summary

Record the final domain and controllers, actual role holders, dated test output, and unresolved issues. Reference files stored in your own environment; do not record credentials.

RMF CONNECTION Implementation records what was configured. Assessment examines evidence. Monitoring revisits conclusions as the system changes.

Completion checklist

☐ I recorded actual settings and the reasons for them.

☐ I checked both controllers' health and replication results.

☐ I kept dated evidence and described failures or limits.

☐ I listed recovery, hardening, and client checks still to do.

What did you observe

Evidence file names, collection time, key results, and the next action

GO FURTHER Explain why replication cannot recover every unwanted change. Identify a change that would trigger a new replication or health check.

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Continue the learning path

The workbook and interactive companion use the same five chapters. The recorded walkthrough is coming soon. Read the current sources before adapting the sample to another environment. Next in the path: course 1.2, Configure DNS and time.

Technical guidance

Install Active Directory Domain Services

https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/deploy/install-active-directory-domain-services--level-100-

Diagnose Active Directory replication failures

https://learn.microsoft.com/en-us/troubleshoot/windows-server/active-directory/diagnose-replication-failures

DCDiag verification

https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/dcdiag

Operations master role placement

https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/plan/planning-operations-master-role-placement

Service overview and network port requirements

https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/service-overview-and-network-port-requirements

Risk and control guidance

Risk Management Framework SP 800-37 Rev. 2

https://csrc.nist.gov/pubs/sp/800/37/r2/final

Security controls SP 800-53 Rev. 5

https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final

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